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Deliverable 3.1

'Workflow for monetary claim procedure'

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DEUCE Consortium Disclaimer

The DEUCE consortium consists of the following partners:

No.	Role	Short Name	Legal Name	Country
1	COO	VUB	VRIJE UNIVERSITEIT BRUSSEL	BE
2	BEN	UL	UNIVERZA V LJUBLJANI	SI
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1. Background to the DEUCE project

DEUCE will rely on the successful results obtained within the previous three projects managed by the Consortium: SCAN (2018-2020, JUST-JCOO-AG-2017), EFFORTS (2020-2022, JUST-JCOO-AG-2019) and SCAN II (2018-2021, JUST-JCOO-AG-2017). In its activities, DEUCE will significantly benefit from SCAN II and EFFORTS databases (the former for ESCP Regulation and the latter for EEO and EOP guidelines), from SCAN experience for what concerns surveys and interviews for the creation of the comprehensive Roadmap as well as from the strong and supportive network of various stakeholders (i.e., ECC-Nets, legal practitioners, academics). From a methodological perspective, the synergic collaboration among the legal unit partners in conducting a plethora of research activities and hands-on tasks is demonstrated by the successful outputs achieved in the above-mentioned projects. Furthermore, DEUCE's objectives complement and support the EU-LISA widespread efforts to digitalize EU Justice. Currently the front-runner Agency in managing large-scale IT systems, EU-LISA's core activity consists of interoperability, namely the efficient communication and cooperation of IT systems to provide EU law enforcement officials with quicker access to comprehensive information and to improve cross border judicial cooperation. Hence, by promoting the use of EEO and EOP procedures, which offer simplified instruments for the recovery of monetary claims, DEUCE will enable the development of a favorable environment for the large-scale adoption of the e-CODEX Project, now being incorporated into the EU-LISA framework. The envisaged result thus consists of the implementation of an interoperable, secure and decentralised communication network between national IT systems in cross-border civil and criminal proceedings.

2. Workflow for Monetary claims resolution (MCR)

Monetary claims resolution (MCR) constitutes a pivotal asset of the European juridical framework's judicial and administrative processes. In recent years, ensuring efficient and transparent claims management has become critical for maintaining and facilitating economic transactions and upholding the rights of creditors and debtors since economic activities expand across borders, raising the demand for a streamlined and effective resolution mechanism. In this bargain, to address the complexities of cross-border monetary disputes, the European Union (EU) has established specific legal instruments, such as the European Payment Order (EPO) and the European Enforcement Order (EEO), which simultaneously accelerate and harmonise the resolution of financial claims and simplifying the procedures for legal clarity to claimants and debtors.

The adoption of an ad-hoc workflow management system becomes imperative to mitigate these inefficiencies, which are provided by the involvement of multiple jurisdictions, variations in procedural requirements, and the reliance on traditional paper-based or manual processes. Such systems seamlessly automate and optimise the monetary claim resolution, reducing the manual workload and avoiding relative compliance risks, ensuring that all necessary procedural steps—from the initial filing of a claim to final enforcement and execution—are executed systematically in a controlled and transparent manner. Specifically, a workflow-driven approach could enhance all the highlighted aspects by integrating rule-based processing, which guarantees that each step adheres to legal and procedural standards. Furthermore, real-time tracking, automated notifications, and digital case management functionalities provide legal practitioners and administrative bodies with a more structured and efficient way to handle claims. We analyse different solutions for the implementation of workflow belonging to the Juridical Domain. After thoroughly analysing the features and performance of several systems on the market and in the literature, we decided to use the Camunda framework (https://camunda.com/). This work comprehends a thorough analysis of the monetary claim resolution process, deepening the capabilities and advantages of the Camunda framework and outlining a structured workflow model tailored for handling monetary claims efficiently to address key procedural pain points while ensuring compliance, automation, and scalability and overall reliability of the monetary claim resolution process.

2.1.1 Workflow Management System (WMS)

In modern business contexts, where the need to manage complex, repetitive, and interdependent processes is increasingly evident and critical, new tools are used to address workflow challenges. Workflow Management Systems (WMS) were created to support the needs of organisations that must manage increasingly fast, dynamic, and globalised processes, where expectations for efficiency, quality, and speed are extremely high. Workflow is an organisational model that helps simplify, automate, and optimise the management of repeatable business flows, reducing errors and identifying areas for improvement to maximise efficiency, develop innovative strategies, and grow the business. The term "workflow" refers to the automated and coordinated organisation, according to predefined models, of tasks and people involved in a work process. Each business process consists of a sequence of activities, called tasks, that must be completed to achieve a predetermined goal and move to the next phase. Workflow management defines the organisation and management of projects, optimising workflows by creating repeatable and defined models that increase transparency and control. Workflow software represents the fundamental tool for managing the digital revolution and making remote collaboration between people easier.

It is clear that interaction between parties and rapid transmission of information allows for the effective and efficient execution of work processes, enabling the company to achieve predefined economic goals, reduce costs, and respond quickly to market changes.

In practical terms, workflow tools are software that defines a work process by identifying the stages to be completed, breaking them down into activities and operations to be assigned to the people involved in the project, who then receive accurate and timely information to carry out their assigned tasks, optimising time and costs.

2.1.2 Benefits of WMS

Workflow management defines the organisation and management of activities and processes within a company by dividing work into specific tasks and assigning them to individuals or teams based on their responsibilities. Through workflows, the methods of executing activities are specified, tasks are assigned across the various stages of the process, and active progress monitoring is conducted. Workflow management, born to reorganising human resources, automates processes through IT systems, enabling the definition, management, and optimisation of workflows automatically, without the need for continuous manual

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intervention. It automates repetitive tasks, assigns duties and priorities, and monitors the progress of each phase of the process.
Numerous benefits are associated with using WFMS, as shown in the figure **Fig. 1**.

1. Simplification and speeding up of processes
2. Reduction of the risk of errors and the need for rework
3. Increased productivity
4. Real-time data and information tracking
5. Clear definition of group and individual performance
6. Identification and removal of slowdowns in the production process
7. Greater transparency in communications
8. Distribution of work according to skills

Fig. 1 Benefits of WMS

2.1.3 WMS in Legal Domain

Workflow management systems (WMS) are very useful and widely used for efficient workflow management in various sectors, and they play an important role at the legal level as well, offering numerous advantages in managing legal and regulatory processes, as shown in the Table 1.

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Advantages of WMS in the Legal Domain
1. Improved Timeliness and Reduction of Bottlenecks
2. Document Management and Secure Archiving
3. Compliance with Regulations and Certified Documentation
4. Facilitating Collaboration Among Legal Professionals
5. Management and Archiving of the History of Each Case
6. Automation and Optimization of Repetitive Tasks

Tab.1 Benefits of WMS in the Legal Domain

Legal practices and workflows are complex, requiring accurate and timely management of documents and data, and must comply with specific regulations to protect individuals' privacy.

Workflow management systems (WMS) assist in managing tasks and processes at the legal level, ensuring that all process stages are followed correctly and in compliance with the law. They automate the document approval flow, reducing the risk of errors and ensuring that all practices are managed in compliance with regulations such as GDPR, tax regulations, or other specific rules. WMS automates legal documents' approval flow and management, ensuring privacy protection and data security. Moreover, processes must be timely and meet specific deadlines, such as those required by contracts, courts, or other legal institutions. Therefore, WMS allows for real-time monitoring of workflows, identifying any delays or bottlenecks in the process, and intervening promptly to keep things on schedule.

Legal processes often involve multiple roles, such as lawyers, legal consultants, and assistants. Therefore, a workflow management system facilitates collaboration among legal teams, allowing them to share documents, information, and case updates in real-time. A WMS allows for the automation of many tasks, reducing manual workload and the risk of errors while improving productivity.

In the legal sector, many tasks are repetitive and standardisable, such as drafting contracts, verifying legal documents, or preparing reports. Key aspects of great importance are the transparency and accountability of processes, which are ensured through data traceability, document version management, and the ability to monitor progress. WMS allow for maintaining a detailed history of each legal case, including steps, approvals, document changes, communications, and decisions made during the process, thus facilitating daily work and providing an important resource for legal defence in case of future disputes or audits. Automating and digitalising workflows reduce the possibility of errors related to manual document management or verbal communication, increasing accuracy and preventing misunderstandings, omissions, or mistakes that could compromise the outcome of a legal case.

2.2 Business Process Management (BPM)

Business Process Management is a discipline that focuses on analysing, improving, automating, and monitoring business processes within an organisation to improve efficiency, reduce costs, increase productivity, and optimise workflows. BPM is an approach that enables companies to manage their processes more effectively and achieve better results.

BPM manages business processes through a continuous cycle of design, modelling, analysis and execution, optimisation, and monitoring to improve efficiency and achieve organisational goals Fig. 2. It involves adopting methodologies, tools, and technologies to ensure that business processes are executed in the most efficient way possible, with continuous improvement aligned with business objectives. Additionally, workflow automation software, such as Workflow Management Systems (WMS), can automate many business processes.

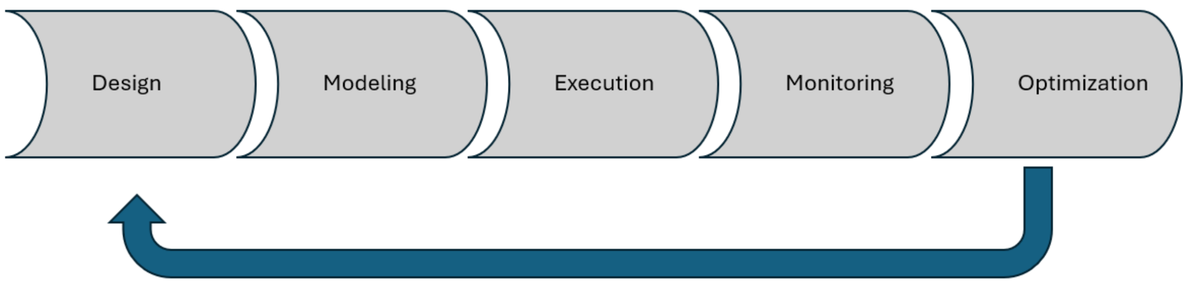

Fig. 2 BPM Cycle

2.2.1. Business Process Model and Notation (BPMN)

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BPMN is a standard used to model and graphically represent workflows, making it easier to understand and automate business processes. Business Process Model and Notation 2.0 (BPMN) is an industrial standard for modelling and executing business processes. A BPMN process is represented by an XML document that includes a visual representation. This duality makes BPMN very powerful: the XML document contains all the information necessary to be interpreted by workflow engines; the visual representation provides enough information to be easily understood by non-technical people.

Sequence Flow
A fundamental concept in BPMN is the sequence flow, which defines the order in which the stages of a process occur. In the visual representation of BPMN, the sequence flow is an arrow that connects two elements. The direction of the arrow indicates the order of execution.

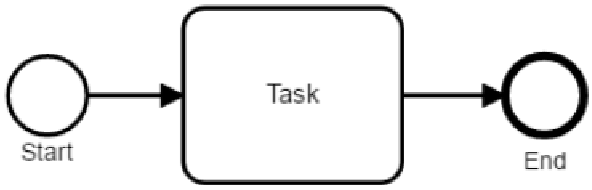

Fig. 3 BPMN Atomic Task formalization

2.3 The Workflow Management Framework

To properly model the domain stream of information, we analysed different solutions for implementing workflows belonging to the Juridical Domain.

After thoroughly analysing the features and performance of several systems on the market and in the literature, we selected the Camunda framework (https://camunda.com/). One of the most advanced business process management (BPM) frameworks that can be leveraged for workflow implementation is Camunda.

This open-source framework enables the design, execution, and monitoring of complex business processes. It is well-suited for managing legal and administrative workflows in monetary claim resolution. Camunada facilitates seamless integration with existing case management systems, digital document repositories, and electronic communication channels, ensuring a fully digitalised and optimised resolution process.
More in detail, Camunda is a process automation platform that helps organisations design, automate, and improve their business processes and workflows. It provides tools and frameworks centred around BPMN (Business Process Model and Notation) and DMN (Decision Model and Notation) standards, allowing developers and business users to collaborate on process design and execution.

Camunda is characterised by several key components, like Process and Decision Modeling, which enable users to visually model business processes using BPMN.
It also supports decision modelling with DMN, allowing the separation of business rules from process flows.

Once a process is modelled, it can be deployed to the Camunda engine. The engine orchestrates tasks, connects to external systems (e.g., via REST APIs), and monitors the workflow state in real-time.

Camunda is lightweight and can be integrated with various technology stacks, such as Java, Spring Boot, microservices, and more. It offers REST APIs and client libraries, making it relatively straightforward to integrate into new or existing systems.
Camunda can handle many automation scenarios, from small teams to large-scale enterprise deployments. It is designed to be horizontally scalable, allowing organisations to process high volumes of tasks without performance bottlenecks.
Camunda includes a web application (Camunda Tasklist) where human tasks requiring user input or review can be managed and tracked.

It facilitates assigning, prioritising, and completing tasks within business processes.
The system has Monitoring and Optimisation facilities: the Camunda Cockpit application provides a real-time view of running processes, incidents, and performance metrics in order to allow users to identify bottlenecks and continuously improve processes based on insights from operational data. The key point in choosing this system is the Open Source Core.

Camunda's core engine is open source, making it accessible to many users and communities. Enterprise editions offer added features such as advanced operational tooling, support, and SLAs.
We concluded that Camunda helps bridge the gap between business requirements and IT by providing a standardised approach to process modelling and execution. By aligning people, systems, and data, Camunda enables end-to-end process automation and continuous improvement across an organisation's operations.

3. The European Monetary Claim Resolution Process

The Monetary Claim Resolution (MCR) Process obeys a carefully regulated series of stages, each designed to preserve the rights of both the claimant and the defendant, thereby ensuring compliance with European Union regulations at the country level. This structured pathway facilitates efficiency and upholds transparency and legal certainty throughout the proceedings. Constituting steps are depicted in the following.

Claimant Initiation
The workflow starts with the claimant's application to the European Payment Order (EPO), generally through a standardized form which comprehends information regarding the nature of the claim, the amount in dispute, and the supporting documentation that establishes the basis for the claimant's demands. The produced, formally recognized application ensures that the procedure is set in motion within a valid and official context.

Court Review and Validation
Subsequently, the application is undertaken for a thorough review by the juridical authority to address its accuracy, completeness, and adherence to legal requirements. During this stage, the court verifies whether the claim meets all formal criteria, such as jurisdictional rules, correct use of the standardized form, and sufficient supporting evidence, ensuring, through this meticulous scrutiny, that unfounded or incomplete claims are promptly identified and preventing unwarranted legal proceedings.

Issuance of the European Payment Order
If the claim is judged consistent and carrying applicable regulations, the process proceeds to issue a European Payment Order (EPO), usually within 30 days from the claimant's request. The EPO, which formally acknowledges that the claim has been validated, is subject to the defendant's right to respond or oppose.

Notification to the Defendant
The EPO is communicated to the defendant with this notification that signifies the official moment at which the defendant is made aware of the claim and is provided with a 30-day window to respond. In this way, the system promotes fairness and upholds the principle that both parties must be granted an equal opportunity to present their case.

Defendant's Response
Upon receiving the EPO, the defendant can take one of several possible actions. Should the defendant decide to fulfil the payment obligation in full, the process concludes, and no further legal measures are needed. Alternatively, the defendant may file a statement of opposition if they wish to contest either the validity or the amount of the claim. In such cases, more detailed legal proceedings typically follow, determining if the defendant neither pays nor opposes the EPO within the stipulated timeframe, with the EPO automatically becoming enforceable in the latter case, reflecting the court's implicit recognition of the claimant's demands.

Further Legal Proceedings
When opposition is lodged or when the nature of the dispute necessitates additional legal scrutiny, the matter can progress to one of several pathways with the claimant that may transfer the case to the competent national civil courts for a full trial, allowing for in-depth examination and adjudication. In the other case, where applicable, the claimant might decide to proceed under the European Small Claims Procedure (SCP). In some cases, the claimant

may choose to discontinue the claim altogether, effectively ending the process if they deem it no longer viable or necessary.

Enforcement via the European Enforcement Order

Suppose the judicial proceedings ultimately result in a favourable judgment for the claimant. In that case, the next step is to secure a European Enforcement Order (EEO) enabling the cross-border enforcement of the judgment in other EU member states. The EEO certificate, designating a key instrument ensuring that court decisions are identified, drastically mitigates the complexity of traditional transnational enforcement procedures.

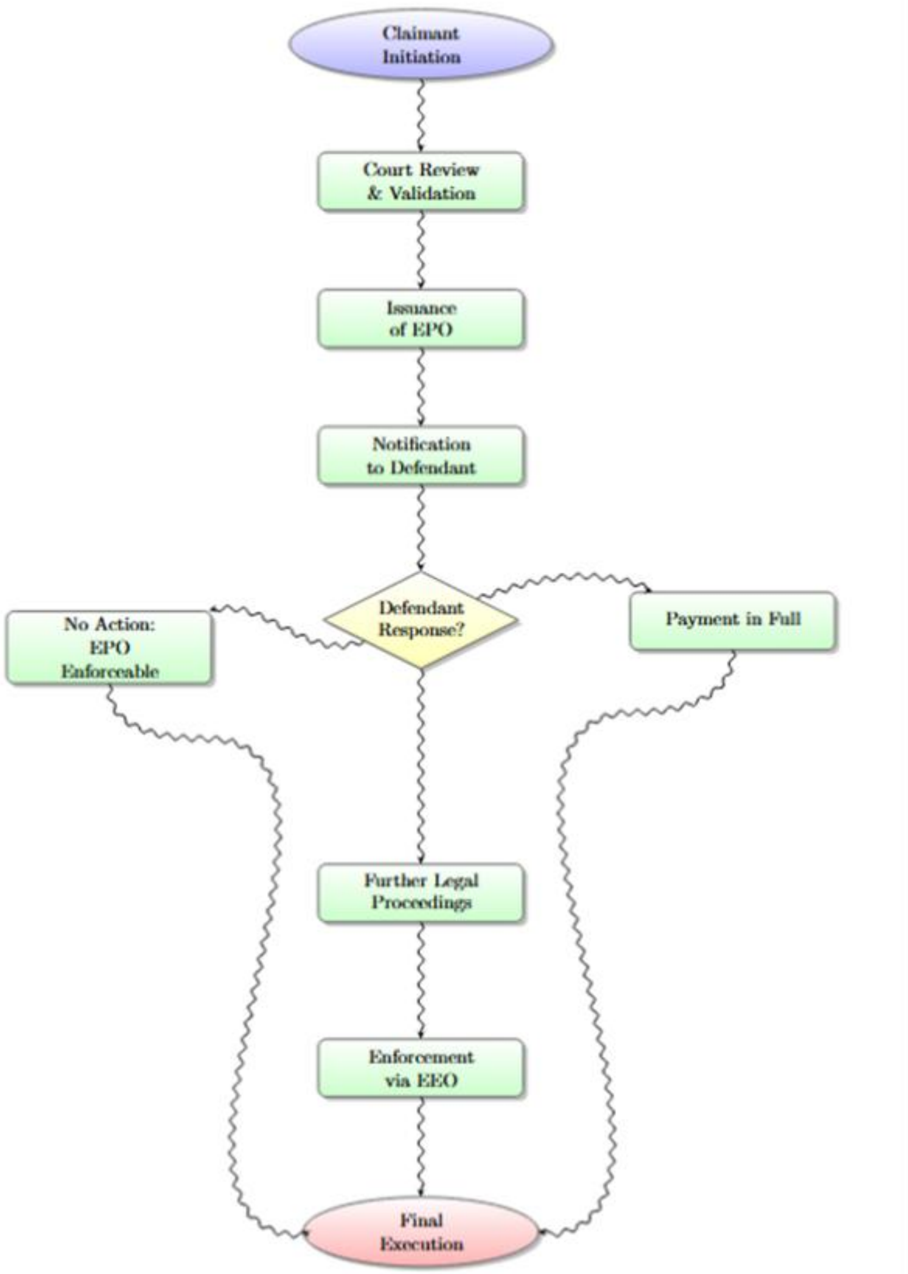

Fig. 4 Monetary Claim Resolution Process

Final Execution and Enforcement

Once the EEO has been obtained and transmitted to the designated enforcement authority within the debtor's jurisdiction, the enforcement authority undertakes the necessary measures to ensure that the judgment is fulfilled—whether by seizing assets, garnishing wages, or taking alternative actions as mandated by law. This final phase closes the loop on the monetary claim resolution process, guaranteeing that successful claimants can rely on an effective mechanism to collect what is owed. From this perspective, adopting an advanced workflow automation platform is crucial for ensuring that complex judicial and administrative procedures—such as the resolution of monetary claims—are handled with maximum efficiency and transparency. Camunda BPM (Business Process Management), an open-source solution that addresses these needs, offering a range of functionalities, significantly reduces procedural bottlenecks while helping, if correctly applied, maintain high compliance and accountability standards.

4. BPM Framework details

The efficient administration of judicial and administration processes can be effectively decomposed into a sequence of concurrent procedural phases. As already said, we selected the tool CAMUNDA BPM as Business Process Management software. It is suitable for processing modelling, is implemented like a free software platform, and is designed specifically to administer complex workflows through features of planning, execution, tracking, and processing. The selected software enables legal professionals and interested parties to streamline processes at no expense of compliance, transparency, and procedural integrity.

4.1 Usage of CAMUNDA BPM in Monetary Claim Management

Applying CAMUNDA BPM for financial claim administration offers significant advantages in procedural efficiency and stakeholder satisfaction; by shifting from traditional manual processes to a digitized, rule-based workflow, judicial and administrative entities can optimize each phase of claim processing. The proposed automation of critical tasks will shorten processing time while significantly enhancing reliability through CRUD operations; moreover, transitioning to a fully digital environment reduces paperwork, streamlines data entry, and minimizes the risk of human errors.

Another equally important benefit is the platform's increased transparency through its real-time tracking capabilities. It enables authorized entities to monitor case progression and access relevant documents at any stage, fostering trust in judicial institutions and ensuring that all stakeholders receive timely and accurate information. CAMUNDA BPM facilitates communication, offering standardized processes and shared data sources, integrating disparate organizations into a cohesive framework and minimizing duplication and oversight while expediting the resolution of contested cases.

4.3 Camunda Framework in Monetary Claim Process

The configuration in Fig.2 positions the Court at the top, signifying its judicial supervision, while CAMUNDA BPM (the Workflow Engine) is centrally placed, reflecting its role in coordinating various processes. At the top-left, the Claimant initiates the process by submitting documents through CAMUNDA BPM, which subsequently advances to the Validation & EPO stage on the top-right. The Notifications & Deadlines and Opposition & Follow-up processes are positioned above CAMUNDA BPM, closer to the Court, emphasizing their regulatory and procedural significance. Below CAMUNDA BPM, the Enforcement Authority receives instructions for final execution. On the far right, the Legal Registry database engages in bidirectional communication with CAMUNDA BPM, facilitating seamless retrieval and updating of claim-related information. This structured configuration ensures that the Workflow Engine effectively integrates all involved entities and procedural steps, promoting efficient task execution, compliance with legal requirements, and sustained transparency throughout the process.

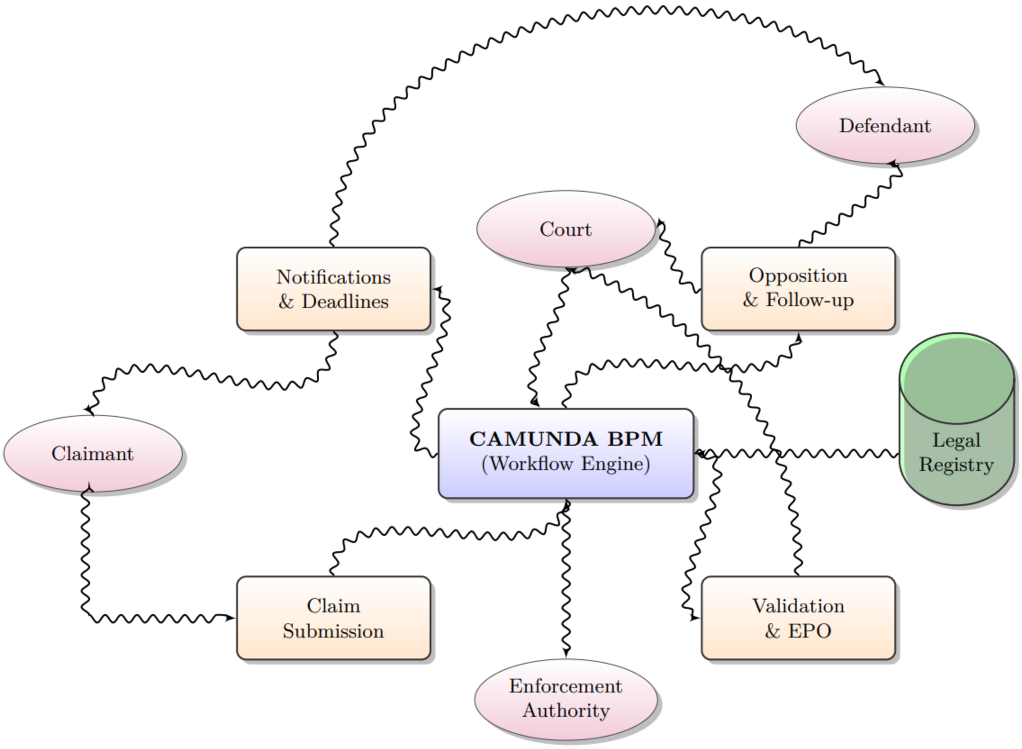

Fig. 5 Camunda Framework in Monetary Claim Process

5. Implementation of the Workflow System

Installing and developing an automated claims resolution process must comprehensively encompass all procedural phases, beginning with the claimant's initial submission and extending through to the final enforcement stage. The subsequent sections will provide a detailed breakdown of the critical workflow components, elucidating their functional significance and representation within the broader process architecture. This structured approach ensures that each procedural step is effectively managed, legal requirements are upheld, and stakeholders are equipped with the necessary tools to navigate and oversee the claims process efficiently.

5.1 Key Workflow Components

The BPM-powered workflow for monetary claim resolution is built upon a series of essential components, each meticulously designed to align with specific operational needs and legal mandates. These components work in unison to establish a seamless digital framework that optimizes every stage of the claims process, from the initial submission and validation to the review, decision-making, and ultimate enforcement of rulings. By leveraging automation, the system minimizes delays, enhances procedural accuracy, and ensures that all actions adhere to regulatory guidelines. Claimants benefit from a structured and transparent process, while legal and administrative bodies gain improved oversight and efficiency in handling cases. Furthermore, the integration of BPM facilitates real-time tracking, communication, and compliance enforcement, ensuring that all involved parties remain informed and engaged throughout the claim lifecycle. Ultimately, this sophisticated digital workflow expedites resolutions and fosters greater confidence in the judicial and administrative processes, reinforcing fairness, accountability, and operational excellence in monetary claim management. In the following, the main components are described.

Digital Submission Portal

The process begins with a secure, web-based platform that allows claimants to electronically submit all required documentation; this digital portal is designed to streamline data collection through standardized forms, ensuring that essential information, such as claimant and debtor details, the disputed amount, and supporting documentation, is captured following both national and European Union regulations. Then, to enhance efficiency, the system helps to prevent leveraging validation features and incomplete or erroneous submissions, minimizing manual errors and lowering administrative workload. Once submitted, the documents become immediately accessible to all the actors involved, expediting the review and validation process and strengthening procedural transparency.

Automated Court Processing

Once a claimant's documents have been submitted, BPM starts a predefined sequence of rules that govern the court's verification and approval procedures, such as determining whether the claim falls within the court's jurisdiction and assessing if it meets the legal thresholds required for an EPO (European Payment Order). Moreover, courts can configure sophisticated internal decision-making rules that automatically route documents and cases to specialized judges or administrators on factors such as case complexity or dispute type, still enhancing efficiency and ensuring that each claim is reviewed by the most appropriate legal authority while maintaining a streamlined and compliant judicial process.

Defendant Notification and Response Tracking

Subsequently, with the issuance of a European Payment Order or a comparable instrument, the BPM triggers a structured notification workflow to inform the defendant of the claim involving the generation of formal notice, translating it if necessary, and ensuring its transmission through legally recognized channels (such as registered mail or secure electronic delivery). Once notified, the defendant's response is closely monitored by the workflow system, which tracks every stage of communication, sets reminders for deadlines (commonly 30 days), and updates the relevant court officials or claimant representatives when a response, be it a full payment, an opposition, or non-action, is logged. This systematic tracking minimizes oversights and helps all parties meet statutory requirements.

Decision Support for Claimants

Where a claim is objected to, a claimant is taken through a series of guided options for alternative legal channels. Options often include having a matter heard in national courts, pursued through the European Small Claims Procedure, or withdrawing a claim. Rule-guided guidance—based on restrictions in terms of procedure, legislative timescales, and relevant EU directives—is then utilized to channel claimants through a most applicable route through user-friendliness that incorporates compliance checking with ease of use, less disorientation and opening judicial access to a broader population.

EEO Enforcement Workflow

The final core component involves automating cross-border enforcement steps by issuing a European Enforcement Order (EEO). When a judgment favouring the claimant is secured, BPM system coordinates the application for an EEO, generates the requisite documents, and synchronizes data with enforcement authorities across multiple EU jurisdictions. This digital integration ensures that judgments are recognized, reducing the delays and complexities that often arise from transnational procedures, thereby maintaining a unified, digital record of each enforcement action and supporting comprehensive tracking and reporting, which are essential for maintaining transparency and accountability.

5.2 Visual Representation of the Workflow

The architectural design of a BPM-driven process is most effectively communicated through a graphical workflow diagram, which highlights the sequential relationship and interdependency of the components. Typically, these diagrams are created using BPMN 2.0 notation and can be segmented into swim lanes or phases to represent the roles played by different actors, such as claimants, courts, and enforcement authorities. In a comprehensive illustrative model, shown in Fig. 3, the "Digital Submission Portal" is positioned at the start of the workflow, with a clear path leading to "Automated Court Processing," signifying the initial validation and issuance of orders. From there, branching paths address the "Defendant Notification and Response Tracking" stage, where timed events and message flows ensure that all parties receive notifications and reminders aligned with statutory deadlines. Once the defendant has responded. This stage is often represented in the diagram by decision gateways or choice nodes, reflecting the claimant's power to select among multiple procedural routes in the event of an opposition. Finally, the process culminates in the "EEO Enforcement Workflow" wherein successful judgments are transmitted to enforcement authorities at the national or EU level.

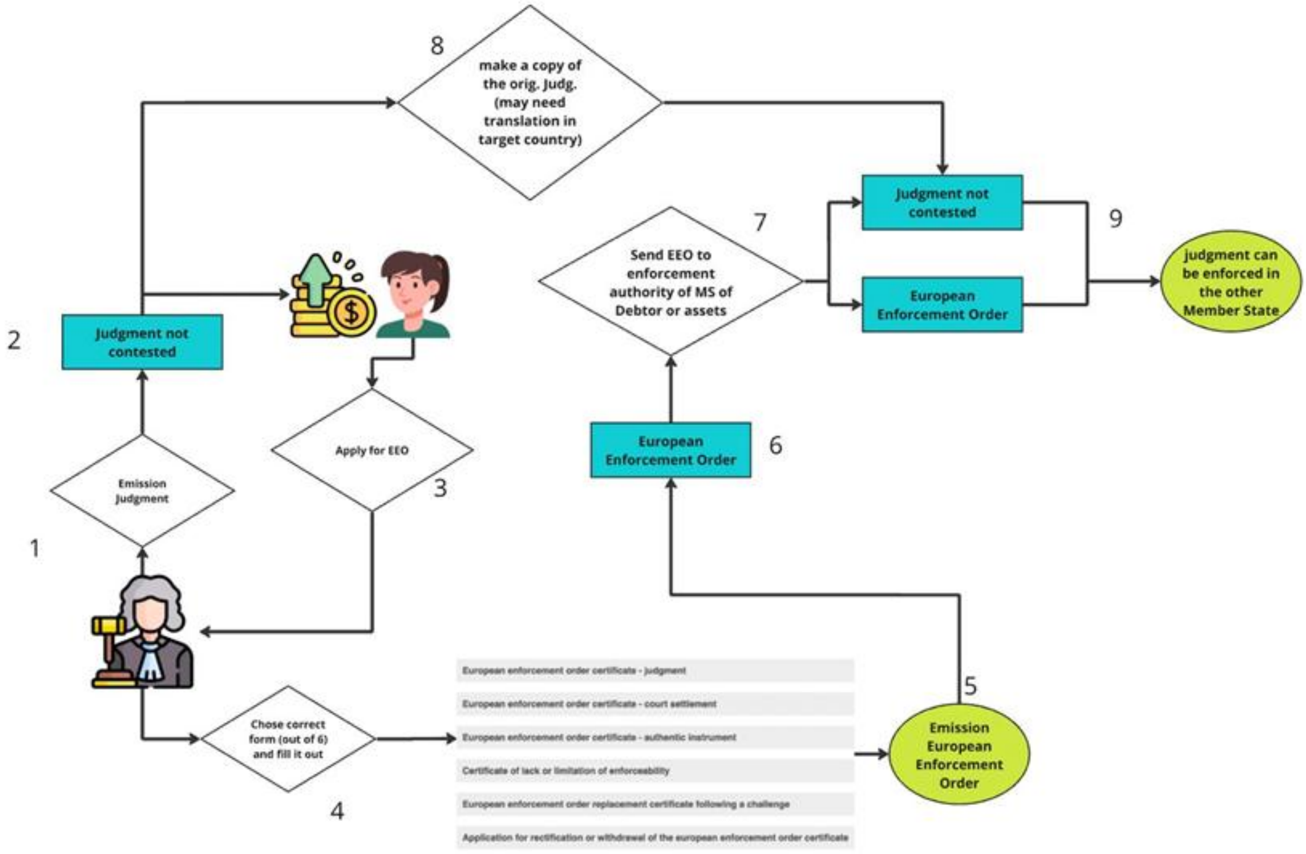

Fig. 6 Information flow

5.3 Workflow for the European Payment Order (EPO) procedure

We deploy a BPMN diagram to effectively map out the European Payment Order (EPO) procedure, detailing how a claim is filed, reviewed, and enforced or disputed. It visually represents how disputes can escalate into a national or small claims process based on the Defendant's actions.

In Fig. 7, The BPMN (Business Process Model and Notation) diagram represents the European Payment Order (EPO) procedure, illustrating the interactions between three key participants: Claimant, Court, and Defendant.
The BPMN Diagram depicted aims to model the following information.

Step 1. Claimant's Actions

The process begins with the Claimant applying for the European Payment Order (EPO).
They fill out Standard Form A and send it to the court.
The process ends if the court rejects the application (Rejection EOP).

Step 2. Court's Processing

The court examines the application for correctness:
If incorrect, the process stops (leading to Rejection EOP).
If correct, the court issues the European Payment Order within 30 days.
The European Payment Order (EPO) is sent to the Defendant.

Step 3. Defendant's Response
The Defendant receives the European Payment Order (EPO) and has three options:
Pay the claim in full → The process ends.
Do nothing → The European Payment Order automatically becomes enforceable.
File a statement of opposition:
This results in the case moving to national courts.

Step 4. Further Court Actions
If the Defendant files a statement of opposition, the court can:
Transfer the case to normal civil courts, where the national court decides based on national law (leading to a Judgment).
Continue the case under the European Small Claims procedure, leading to a Final decision.
Discontinue the procedure, ending the process.

5.4 Workflow for the European Enforcement Order (EOO) procedure

The diagram in Fig.8 represents the European Enforcement Order (EEO) Process, structured as a swimlane diagram that separates responsibilities between The Court and The Applicant.

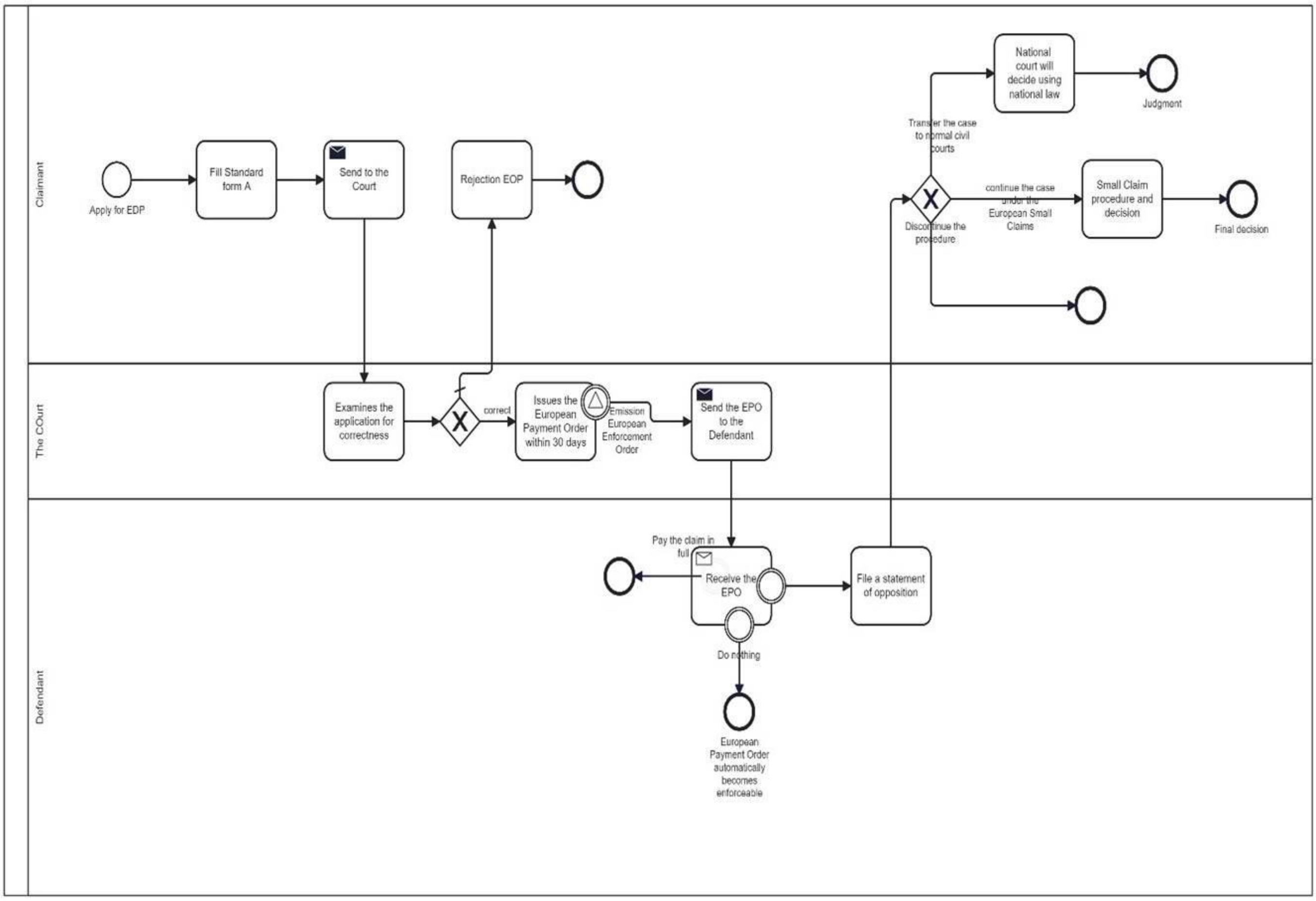

Fig. 7 Detailed EPO-BPMN Workflow

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This workflow represents the **European Enforcement Order (EEO) Process**, structured as a **swimlane diagram** that separates responsibilities between:

- **The Court**: Handles judgment emission, verification, and enforcement certification.
- **The Applicant**: Submits requests for EEO and ensures necessary documentation is completed.
-

Step 1. Emission of Judgement

- The process begins with the **court issuing a judgment**.
- This judgment is a legal decision regarding a dispute.

Step 2. Decision on Contestation

- If the judgment is **not contested**, it can proceed directly to enforcement.
- If it **is contested**, additional legal proceedings may be required (not shown in this diagram).

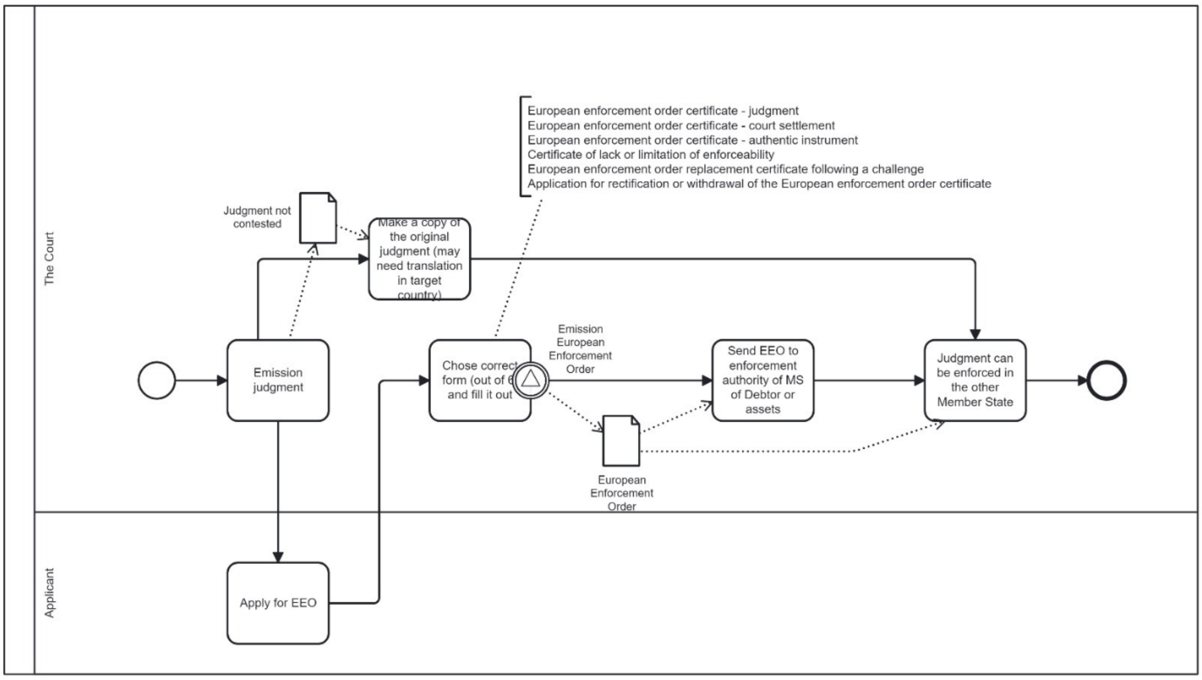

Fig. 8 Detailed EOO-BPMN Workflow

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Step 3. Applying for a European Enforcement Order (EEO)

- If the judgment is **not contested**, the applicant can **apply for an EEO**.
- The **EEO is a certification** that allows the judgment to be recognized and enforced in another EU Member State **without additional legal proceedings**.

Step 4. Form Compilation

The applicant must choose the correct form from the following:

1. European enforcement order certificate - judgment
2. European enforcement order certificate - court settlement
3. European enforcement order certificate - authentic instrument
4. Certificate of lack or limitation of enforceability
5. European enforcement order replacement certificate following a challenge
6. Application for rectification or withdrawal of the European enforcement order certificate

- The correct form must be filled out and submitted for approval.

Step 5. Issuance of European Enforcement Order (EEO)

- Once the correct form is filled, the court issues the European Enforcement Order.
- The EEO is a document certifying that the judgment meets the necessary conditions for enforcement in another EU Member State.

Step 6. Copy and Translation of the Judgment (If Required)

- A copy of the original judgment may be required.
- In some cases, a **translation** might be necessary to comply with the language requirements of the target Member State.

Step 7: Sending the EEO to the Enforcement Authority

- The EEO is sent to the enforcement authority of the Member State, where the debtor or their assets are located.
- This step ensures that the relevant authorities can process the judgment in the target country.

Step 8: Final Enforcement of the Judgment

- Once received, the judgment can now be enforced in the other Member State.
- This means that debt collection or asset seizure can be carried out under the laws of the enforcing country.

In order to model these diagrams, we used the following Key BPMN Elements.
Swimlanes: Represent different entities (Claimant, Court, and Defendant).
Events: Start and end events indicate the initiation and conclusion of the process.
Tasks include filling out the form, examining correctness, and sending EPO.
Decision Gateways: Conditional paths (e.g., correctness check, Defendant's response).
Arrows/Flows: Show the sequence of activities.

6. Conclusion

Implementing a workflow management system using a BPM system represents a transformative step in modernizing the resolution of monetary claims. By replacing traditional manual workflows with digital automation, legal and administrative entities can significantly improve efficiency, transparency, compliance, and case management.

By leveraging BPM, courts, claimants, and enforcement agencies benefit from:

- **Reduced processing times** and faster case resolution.
- **Automated tracking and reporting** for better accountability.
- **Optimized resource allocation**, reducing operational costs.
- **Seamless cross-border enforcement** within EU member states.

In conclusion, adopting the proposed automated, structured, and legally compliant workflow system powered by BPM system ensures that monetary claim resolution is conducted efficiently, fairly, and transparently, reinforcing trust in the legal system and facilitating smoother economic transactions across the European Union.

Appendix A

In the following we report the code of the Workflow for the European Payment Order (EPO) procedure, written in BPMN 2.0 (Business Process Model and Notation), XML representation of the business process model. It defines the workflow that involves multiple tasks, participants (like "The Court" and "Applicant"), events, and data objects.

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xmlns:bpmndi="http://www.omg.org/spec/BPMN/20100524/DI"
xmlns:dc="http://www.omg.org/spec/DD/20100524/DC"
xmlns:di="http://www.omg.org/spec/DD/20100524/DI"
xmlns:modeler="http://camunda.org/schema/modeler/1.0"
id="Definitions_1y1lkdi"
targetNamespace="http://bpmn.io/schema/bpmn"      exporter="Camunda
Modeler" exporterVersion="5.21.0" modeler:executionPlatform="Camunda
Cloud" modeler:executionPlatformVersion="8.4.0">
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processRef="Process_0dz68ag" />
  </bpmn:collaboration>
  <bpmn:process id="Process_0dz68ag" isExecutable="true">
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      <bpmn:lane id="Lane_054icj2" name="Claimant">
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        <bpmn:flowNodeRef>Activity_03lldhg</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_1j0s1ud</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_1u60con</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_1bp11ss</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_16hm3ai</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Gateway_04ew057</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_0sxrm97</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_0m93q4e</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_1vwyl2w</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_07e4fgh</bpmn:flowNodeRef>
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      <bpmn:lane id="Lane_0fmi5fn" name="The Court">
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```

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```
        <bpmn:flowNodeRef>Activity_1aakr3r</bpmn:flowNodeRef>
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        <bpmn:flowNodeRef>Activity_05zlf92</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_1byf0g2</bpmn:flowNodeRef>
      </bpmn:lane>
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        <bpmn:flowNodeRef>Event_0wobzht</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_18u23kq</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_08pn28d</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_12qnrgw</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_0v66h55</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_0yjhraf</bpmn:flowNodeRef>
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    </bpmn:laneSet>
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default="Flow_073aa0j">
      <bpmn:incoming>Flow_0cvp3j4</bpmn:incoming>
      <bpmn:outgoing>Flow_1y1r8am</bpmn:outgoing>
      <bpmn:outgoing>Flow_073aa0j</bpmn:outgoing>
    </bpmn:exclusiveGateway>
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    </bpmn:task>
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    </bpmn:startEvent>
    <bpmn:task id="Activity_03lldhg" name="Fill Standard form A">
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      <bpmn:outgoing>Flow_0rchnxs</bpmn:outgoing>
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    </bpmn:sendTask>
    <bpmn:task id="Activity_1u60con" name="Rejection EOP">
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    </bpmn:task>
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    </bpmn:task>
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decide using national law">
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      <bpmn:outgoing>Flow_18quxsu</bpmn:outgoing>
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      <bpmn:outgoing>Flow_03c5lwh</bpmn:outgoing>
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```

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```
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      <bpmn:outgoing>Flow_1jj63oo</bpmn:outgoing>
    </bpmn:task>
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    </bpmn:endEvent>
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    <bpmn:endEvent id="Event_07e4fgh">
      <bpmn:incoming>Flow_1p1r0oh</bpmn:incoming>
    </bpmn:endEvent>
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    <bpmn:sendTask id="Activity_05zlf92" name="Send the EPO to the
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    </bpmn:sendTask>
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automatically becomes enforceable">
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opposition">
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```

```
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targetRef="Activity_03lldhg" />
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sourceRef="Activity_03lldhg" targetRef="Activity_1j0s1ud" />
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sourceRef="Activity_1j0s1ud" targetRef="Activity_1gn59aw" />
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sourceRef="Activity_16hm3ai" targetRef="Event_1vwyl2w" />
    <bpmn:sequenceFlow id="Flow_1u51zax"
sourceRef="Activity_08pn28d" targetRef="Gateway_04ew057" />
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targetRef="Activity_0sxrm97" />
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bpmnElement="Event_0yjhraf">
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```

```
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      &lt;/bpmndi:BPMNEdge&gt;
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bpmnElement="Flow_0cvp3j4">
        &lt;di:waypoint x="630" y="510" /&gt;
        &lt;di:waypoint x="675" y="510" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_1y1r8am_di"
bpmnElement="Flow_1y1r8am">
        &lt;di:waypoint x="725" y="510" /&gt;
        &lt;di:waypoint x="770" y="510" /&gt;
        &lt;bpmndi:BPMNLabel&gt;
          &lt;dc:Bounds x="730" y="492" width="35" height="14" /&gt;
        &lt;/bpmndi:BPMNLabel&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_0p1ww5w_di"
bpmnElement="Flow_0p1ww5w">
        &lt;di:waypoint x="888" y="490" /&gt;
        &lt;di:waypoint x="939" y="490" /&gt;
        &lt;di:waypoint x="939" y="510" /&gt;
        &lt;di:waypoint x="990" y="510" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_14mr4n8_di"
bpmnElement="Flow_14mr4n8">
        &lt;di:waypoint x="1040" y="550" /&gt;
        &lt;di:waypoint x="1040" y="640" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_073aa0j_di"
bpmnElement="Flow_073aa0j">
        &lt;di:waypoint x="700" y="485" /&gt;
        &lt;di:waypoint x="700" y="460" /&gt;
        &lt;di:waypoint x="740" y="460" /&gt;
        &lt;di:waypoint x="740" y="300" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_1imj8t2_di"
bpmnElement="Flow_1imj8t2">
        &lt;di:waypoint x="1108" y="680" /&gt;
        &lt;di:waypoint x="1170" y="680" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_1u51zax_di"
bpmnElement="Flow_1u51zax">
        &lt;di:waypoint x="1220" y="640" /&gt;
        &lt;di:waypoint x="1220" y="280" /&gt;
        &lt;di:waypoint x="1265" y="280" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_0h6wwsn_di"
bpmnElement="Flow_0h6wwsn">
        &lt;di:waypoint x="972" y="660" /&gt;
        &lt;di:waypoint x="950" y="660" /&gt;
        &lt;di:waypoint x="950" y="670" /&gt;
        &lt;di:waypoint x="918" y="670" /&gt;
      &lt;/bpmndi:BPMNEdge&gt;
      <bpmndi:BPMNEdge                                    id="Flow_03c5lwh_di"
bpmnElement="Flow_03c5lwh">
        &lt;di:waypoint x="1290" y="255" /&gt;
        &lt;di:waypoint x="1290" y="150" /&gt;
```

```
        <di:waypoint x="1370" y="150" />
        <bpmndi:BPMNLabel>
          <dc:Bounds x="1262" y="195" width="86" height="40" />
        </bpmndi:BPMNLabel>
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_1dnbkic_di"
bpmnElement="Flow_1dnbkic">
        <di:waypoint x="1315" y="280" />
        <di:waypoint x="1480" y="280" />
        <bpmndi:BPMNLabel>
          <dc:Bounds x="1359" y="262" width="87" height="53" />
        </bpmndi:BPMNLabel>
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_1p1r0oh_di"
bpmnElement="Flow_1p1r0oh">
        <di:waypoint x="1290" y="305" />
        <di:waypoint x="1290" y="390" />
        <di:waypoint x="1472" y="390" />
        <bpmndi:BPMNLabel>
          <dc:Bounds x="1258" y="302" width="76" height="27" />
        </bpmndi:BPMNLabel>
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_18quxsu_di"
bpmnElement="Flow_18quxsu">
        <di:waypoint x="1470" y="150" />
        <di:waypoint x="1562" y="150" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_1jj63oo_di"
bpmnElement="Flow_1jj63oo">
        <di:waypoint x="1580" y="280" />
        <di:waypoint x="1662" y="280" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_0khvpg3_di"
bpmnElement="Flow_0khvpg3">
        <di:waypoint x="288" y="260" />
        <di:waypoint x="370" y="260" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_0rchnxs_di"
bpmnElement="Flow_0rchnxs">
        <di:waypoint x="470" y="260" />
        <di:waypoint x="530" y="260" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_1v9gigy_di"
bpmnElement="Flow_1v9gigy">
        <di:waypoint x="790" y="260" />
        <di:waypoint x="842" y="260" />
      </bpmndi:BPMNEdge>
      <bpmndi:BPMNEdge                          id="Flow_0u7jrxo_di"
bpmnElement="Flow_0u7jrxo">
        <di:waypoint x="1050" y="738" />
        <di:waypoint x="1050" y="792" />
      </bpmndi:BPMNEdge>
    </bpmndi:BPMNPlane>
  </bpmndi:BPMNDiagram>
</bpmn:definitions>
```


Appendix B

In the following we report the code of the Workflow for the European Enforcement Order (EOO) procedure, written in BPMN 2.0 (Business Process Model and Notation), XML representation of the business process model. It defines the workflow that involves multiple tasks, participants (like "The Court" and "Applicant"), events, and data objects.

```
<?xml version="1.0" encoding="UTF-8"?>
<bpmn:definitions
  xmlns:bpmn="http://www.omg.org/spec/BPMN/20100524/MODEL"
  xmlns:bpmndi="http://www.omg.org/spec/BPMN/20100524/DI"
  xmlns:dc="http://www.omg.org/spec/DD/20100524/DC"
  xmlns:di="http://www.omg.org/spec/DD/20100524/DI"
  xmlns:modeler="http://camunda.org/schema/modeler/1.0"
  id="Definitions_1cqfjrs"
  targetNamespace="http://bpmn.io/schema/bpmn"
  exporter="Camunda Modeler"
  exporterVersion="5.21.0"
  modeler:executionPlatform="Camunda Cloud"
  modeler:executionPlatformVersion="8.4.0">

  <bpmn:collaboration id="Collaboration_0iyksx1">
    <bpmn:participant                      id="Participant_0oera4f"
processRef="Process_02m0bj6" />
    <bpmn:textAnnotation id="TextAnnotation_03f2j8v">
      <bpmn:text>
        European enforcement order certificate - judgment
        European enforcement order certificate - court settlement
        European enforcement order certificate - authentic instrument
        Certificate of lack or limitation of enforceability
        European enforcement order replacement certificate following
a challenge
        Application for rectification or withdrawal of the European
enforcement order certificate
      </bpmn:text>
    </bpmn:textAnnotation>
    <bpmn:association id="Association_0lt0jep"
      associationDirection="None"
      sourceRef="Activity_1bgxrwt"
      targetRef="TextAnnotation_03f2j8v" />
  </bpmn:collaboration>

  <bpmn:process id="Process_02m0bj6" isExecutable="true">
    <bpmn:laneSet id="LaneSet_1eg3d4p">
      <bpmn:lane id="Lane_0iw05f8" name="The Court">
        <bpmn:flowNodeRef>StartEvent_1</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_0os7uv8</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_1bgxrwt</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_0i55ric</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_10kf0s2</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_0k4vcji</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Activity_0da47hh</bpmn:flowNodeRef>
        <bpmn:flowNodeRef>Event_1eu69rl</bpmn:flowNodeRef>
      </bpmn:lane>
      <bpmn:lane id="Lane_1dbc7rg" name="Applicant">
        <bpmn:flowNodeRef>Activity_0irabtj</bpmn:flowNodeRef>
      </bpmn:lane>
    </bpmn:laneSet>

    <bpmn:startEvent id="StartEvent_1">
      <bpmn:outgoing>Flow_12nuubk</bpmn:outgoing>
    </bpmn:startEvent>

    <bpmn:task id="Activity_0os7uv8" name="Emission judgment">
```

```
      <bpmn:incoming>Flow_12nuubk</bpmn:incoming>
      <bpmn:outgoing>Flow_1gvrsns</bpmn:outgoing>
      <bpmn:outgoing>Flow_11a1zbq</bpmn:outgoing>
      <bpmn:dataOutputAssociation
id="DataOutputAssociation_0vgetlz">
        <bpmn:targetRef>DataObjectReference_07elzs8</bpmn:targetRef>
      </bpmn:dataOutputAssociation>
    </bpmn:task>

    <bpmn:task id="Activity_1bgxrwt" name="Chose correct form (out of
6) and fill it out">
      <bpmn:incoming>Flow_0eairob</bpmn:incoming>
    </bpmn:task>

    <bpmn:task id="Activity_0i55ric" name="Send EEO to enforcement
authority of MS of Debtor or assets">
      <bpmn:incoming>Flow_15z2eru</bpmn:incoming>
      <bpmn:outgoing>Flow_0ac1z38</bpmn:outgoing>
    </bpmn:task>

    <bpmn:endEvent id="Event_1eu69rl">
      <bpmn:incoming>Flow_0dpvb9t</bpmn:incoming>
    </bpmn:endEvent>
  </bpmn:process>
</bpmn:definitions>
```